Reviewer Pack — Minimal Quadratic Defect and Communication Complexity

Entry e895b1ba0efb · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Quadratic Defect and Communication Complexity

Entry ID: `e895b1ba0efb` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-07 07:38:32 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Quadratic Forms) **Field B** (complexity object): Communication Complexity (Boolean Functions)

Statement:

> For every boolean function f: {0,1}^n → {0,1}, the communication complexity of f, denoted as CC(f), is upper-bounded by the minimal quadratic defect, denoted as D_q(f), defined as the minimum value of |D[f(x_1,...,x_k)] - 1|/k over all distinct inputs x_1,...,x_k ∈ {0,1}^n, where D[f(x_1,...,x_k)] is the quadratic form associated with the boolean function evaluated at (x_1,...,x_k).

Rationale (proposer's reasoning):

> Quadratic forms have been studied in algebraic geometry and have applications in coding theory. The quadratic defect captures the distance of a boolean function from being linear. Communication complexity measures the amount of communication needed to compute a boolean function. This conjecture explores whether the geometric property of quadratic forms can provide insights into the complexity of computing Boolean functions, which is a fundamental problem in complexity theory.

Taxonomy category: `AlgebraicGeometry-CommunicationComplexity` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `fd83420e974677cb`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all generated boolean functions f: {0,1}^n → {0,1}, with n ≤ 40 and 30 random seeds, the minimal quadratic defect D_q(f) equals the communication complexity CC(f) within a tolerance of ±10%.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: `ADJACENT_OK` against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"quadratic form" AND "communication complexity"` - `"minimal quadratic defect" AND boolean function"` - `"algebraic geometry" AND communication complexity`

Top relevant hits considered: - [http://arxiv.org/abs/1409.1534v1] Algorithms in Real Algebraic Geometry: A Survey - [http://arxiv.org/abs/1008.0601v3] A note on the cotangent complex in derived algebraic geometry - [http://arxiv.org/abs/math/0504259v1] Multiplier ideal sheaves in complex and algebraic geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.1s

5.1 Generated Python source

```python
import random
import math
from itertools import combinations

def generate_random_boolean_function(n):
    return {i: random.randint(0, 1) for i in range(2**n)}

def evaluate_quadratic_form(f, x_k):
    n = int(math.log2(len(x_k)))
    Q = [[0] * n for _ in range(n)]
    for i in range(n):
        for j in range(i, n):
            Q[i][j] = f[(i << (n - 1)) | (j << (n - 2))]
            if i != j:
                Q[j][i] = Q[i][j]
    return sum(Q[i][j] * x_k[i] * x_k[j] for i in range(n) for j in range(i, n))

def minimal_quadratic_defect(f):
    n = int(math.log2(len(f)))
    min_defect = float('inf')
    for k in range(1, len(f)):
        for x_k in combinations(range(2**n), k):
            defect = abs(evaluate_quadratic_form(f, x_k) - 1) / len(x_k)
            if defect < min_defect:
                min_defect = defect
    return min_defect

def communication_complexity(f):
    n = int(math.log2(len(f)))
    max_comm_cost = float('-inf')
```

```
    for k in range(1, len(f)):
        for x_k in combinations(range(2**n), k):
            comm_cost = sum(abs(f[i] - f[j]) for i, j in combinations(x_k, 2))
            if comm_cost > max_comm_cost:
                max_comm_cost = comm_cost
    return max_comm_cost

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        f = generate_random_boolean_function(n)
        cc = communication_complexity(f)
        min_defect = minimal_quadratic_defect(f)

        if min_defect == float('inf'):
            return {
                "metric_name": "minimal_quadratic_defect",
                "metric_value": None,
                "instances_tested": 0,
                "n_max": n,
                "conjecture_holds": False,
                "counterexample": "mapping_undefined"
            }

        results.append({
            "n": n,
            "cc": cc,
            "min_defect": min_defect
        })

    mean_cc = sum(result["cc"] for result in results) / len(results)
    mean_min_defect = sum(result["min_defect"] for result in results) / len(results)
    std_dev = math.sqrt(sum((result["cc"] - mean_cc)**2 + (result["min_defect"] - mean_min_defect)

    support_fraction = sum(1 for result in results if abs(result["cc"] - result["min_defect"]) <=

    return {
        "metric_name": "minimal_quadratic_defect",
        "metric_value": mean_min_defect,
        "instances_tested": sum(1 for result in results),
        "n_max": max(result["n"] for result in results),
        "conjecture_holds": support_fraction >= 0.8,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2**i - 1 for i in range(5, 31)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
```

```
    results.append(result)

mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
std_dev = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results) / len(res
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_dev} support_fraction={support
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_dev} support_fraction={support
else:
    first_failing_seed = next(seed for seed, r in zip(seeds, results) if not r["conjecture_hol
    print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test timed out before producing data, which means it did not complete within the allotted time frame. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	23265
2	propose	ollama_remote	glm4:latest	0	0	12989
3	propose	ollama_remote	glm4:latest	0	0	13696
4	preregistration	ollama_remote	glm4:latest	0	0	9692
5	novelty	ollama_remote	glm4:latest	0	0	16551
6	novelty	ollama_remote	glm4:latest	0	0	9998
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	49713
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12104
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11488
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12604
11	judge	ollama_remote	glm4:latest	0	0	28976

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 201077 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/e895b1ba0efb.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e895b1ba0efb.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/e895b1ba0efb.tar.gz` (if generated)